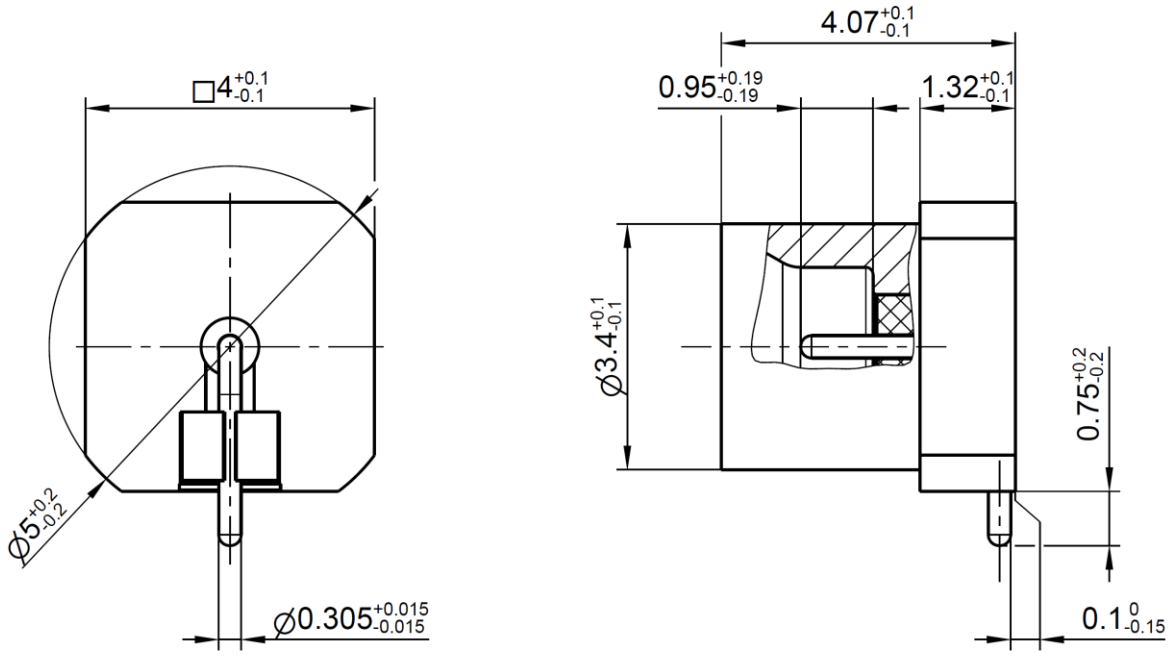
Technical Data Sheet		Rosenberger
Mini-SMP	Straight Plug PCB Smooth Bore	**18S142-40ML5**



All dimensions are in mm

Interface

According to	MIL-STD-348 Mateable with GPPO™ (Gilbert Engineering Co., Inc.) and SSMP™ (Connectors Devices, Inc.)

Documents

PCB layout	B 209
Tape & reel packaging	VG45.1M500

Material and plating

Connector parts	**Material**	**Plating**
Center contact	CuBe	AuroDur®, gold plated
Outer contact	Brass	AuroDur®, gold plated
Dielectric	PEEK	

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Electrical data

Impedance	50 Ω
Frequency	DC to 65 GHz
Return loss	≥ 19 dB, DC to 26.5 GHz ≥ 13 dB, 26.5 to 40 GHz ≥ 9 dB, 40 to 65 GHz
Insertion loss	$\leq 0.05 \times \sqrt{f(GHz)}$ dB
Insulation resistance	≥ 5 GΩ
Center contact resistance	≤ 6.0 mΩ
Outer contact resistance	≤ 2.0 mΩ
Working voltage (at sea level) (at 70000 feet)	325 V rms 125 V rms

- VSWR in application depends decisive on PCB layout -

Mechanical data

Mating cycles	≥ 100
Center contact captivation	≥ 5 N
Engagement force	
- smooth bore	11 N typical
Disengagement force	
- smooth bore	11 N typical

Environmental data

Temperature range	-55°C to +155°C
Thermal shock	MIL-STD-202, Method 107, Condition B
Vibration	MIL-STD-202, Method 204, Condition A
Shock	MIL-STD-202, Method 213, Condition A
Moisture resistance	MIL-STD-202, Method 106
Climatic Category	IEC 60068 55/155/21
Max. soldering temperature	IEC 61760-1, +260°C for 10 sec.
RoHS	compliant

Tooling

N/A

Suitable cables

N/A

Weight

Weight	0.3 g/pce

While the information has been carefully compiled to the best of our knowledge, nothing is intended as representation or warranty on our part and no statement herein shall be construed as recommendation to infringe existing patents. In the effort to improve our products, we reserve the right to make changes judged to be necessary.

For the installation of the electrotechnical equipment, particular electrotechnical expertise is required.

Draft	Date	Approved	Date
S. Andorfer	01.04.25	H. Babinger	01.04.25

Rev.	Engineering change number	Name	Date
g01	25-0545	T. Börgerding	01.04.25

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